claude-windows / 6a82308f-69d6-43a8-9ea4-d8eaf497253d

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\subagents\workflows\wf_3d536535-241\agent-a9b2f9908af4627f6.jsonl`
- SHA-256: `548b190d64af7bc53dfe4f8f48cf07c2f1195585a097bc2e2da87d4bd92dfd18`
- Source modified: `2026-06-16T03:26:43+00:00`
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- project: `wf_3d536535-241`
- session_id: `6a82308f-69d6-43a8-9ea4-d8eaf497253d`

Transcript

1. user (2026-06-16T03:25:56.150Z)

Adversarial Claim Verifier (voter 1/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

Research question

Research question: **What concrete, recent (2022?2025) techniques and evaluation practices should a local, commercially-licensed badminton rally-detector adopt to close the quality gap with a strong cloud VLM (Gemini), given a tiny labeled set (~6 ? ~16 golden videos)?** Produce an adversarially-verified, cited report organized around the four dimensions below. Prefer specific named methods, papers (arXiv/venue), and reported numbers over generic advice; flag anything you cannot verify.

CONTEXT (so you target the real gap, not generic ML):
- System: a local pipeline detects the shuttle per-frame (WASB / TrackNet, MIT-licensed) ? a heuristic turns the trajectory into candidate rally windows ? today an optional Gemini "handoff" confirms rallies. We just measured a fully-local (no-AI) run vs Gemini on 3 matches: local emits far FEWER but MUCH LONGER windows (mean 65s vs 6.1s; one window spanned a whole 174s clip) ? it does NOT miss activity, it FAILS TO SEGMENT. Root cause (code-confirmed): a frame is "active" iff the shuttle is visible AND moving above a small per-second-normalized velocity threshold; active frames within a 2 s gap are merged; there is NO max-window cap and NO inactivity/boundary split. So "shuttle is moving" is conflated with "rally in progress," and dead-time + serves + knock-ups bridge rallies into mega-windows. Shuttle tracking itself is excellent (96?99% per-frame visibility).
- HARD CONSTRAINTS: weights/data/code must be MIT/Apache-2.0/BSD only (no viral/NC/custom licenses, reject Gemma-3/Llama-4-MAU-cap); we may NEVER train owned weights on paid-LLM (Gemini/OpenAI/Anthropic) outputs (terms/copyright), though paid LLMs may be used at inference; minors excluded from any training pool; runs on a single local GPU.
- WHAT WE ALREADY HAVE (do NOT re-survey the basics ? go deeper/newer): the architecture skeleton is known to be standard Temporal Action Localization/Segmentation (proposal+scorer) + late/score-level fusion + stacked generalization; in-domain prior art (MonoTrack, ShuttleSet, See et al. 2024/25 non-broadcast badminton rally start/end, Ghosh et al. point-segmentation, tennis/TT play-break state machines) is already cited. We have a working eval harness: leave-one-video-out (grouped by match), temporal-IoU F1 @0.5, F1@{0.1,0.25,0.5}, mAP@tIoU, segment-count-ratio, bootstrap 95% CIs, paired exact sign-test, Platt/isotonic calibration + Brier/ECE, a tiny numpy LogisticRegression scorer over per-window feature columns, an experiment leaderboard, and an "eval?serve" guard (a cue measured +0.074 F1 on permissive proposal windowing but ?0.008 on the coarser served windowing ? reverted). Available but default-OFF per-window signals: net-crossing count (rally gate), 5 person/court features, 3 audio-hit features.

THE FOUR DIMENSIONS:

1) WINDOWING & BOUNDARIES (Temporal Action Localization/Segmentation): beyond proposal+score ? what are the best *recent* methods for (a) turning a dense per-frame signal into well-segmented

actions, (b) explicitly controlling OVER-segmentation/merging, (c) boundary detection/regression and splitting merged segments? Cover e.g. MS-TCN/MS-TCN++, ASRF (action-boundary regression), BMN/BSN boundary-matching, ActionFormer/TriDet/TemporalMaxer (actionness + boundary heads), smoothing/boundary losses, and the standard metrics for over-segmentation (segmental F1@k, edit/Levenshtein score, mAP@tIoU). Which transfer to a 1-D shuttle-trajectory + auxiliary-cue setting on tiny data?

2) SHUTTLE-TRAJECTORY ? RALLY SIGNAL PROCESSING (racket-sports specific): how do published systems convert ball/shuttle tracking into rally/point segments and distinguish "in-play" from dead-time? Cover rally-state machines, serve/let detection, net-crossing & two-sided-occupancy cues, hit/contact (audio + kinematic) detection, and fps-invariant trajectory features. What features best separate "rally" from "shuttle merely moving"?

3) SMALL-DATA STRATEGIES: with ~6?16 labeled videos, what gives the most quality per label ? active learning (which videos/segments to label next; uncertainty/diversity/core-set), semi-/weakly-/point-supervised TAL (e.g. SF-Net, timestamp supervision), self-training/pseudo-labeling, and data-efficient TAL? Separately: the legally-clean ways to exploit a strong teacher (Gemini) at INFERENCE for weak/pseudo-labels or as an evaluation oracle WITHOUT its outputs becoming training data for owned weights ? what does the literature say about teacher-student / distillation boundaries and label-noise handling, and what governance distinctions matter?

4) EVALUATION METHODOLOGY & RIGOR at small n: trustworthy measurement with ~6?16 videos ? bootstrap/permutation tests, paired sign-tests, leave-one-group-out vs RLOOCV bias, calibration (reliability, ECE), the over-segmentation-blindness of plain tIoU, shadow/A-B evaluation, regression gates, and concrete pitfalls (label noise, annotator agreement/IAA, multiple-comparison/over-fitting traps, eval-vs-serve drift). What metrics + protocols should a small-data sports-segmentation leaderboard standardize on?

Deliverable: a structured, cited report with, per dimension, the 3?5 highest-leverage techniques that APPLY to our constrained setting (small data, local, license-clean, 1-D trajectory + cues + learned scorer), each with: what it is, why it fits our over-merge/small-data problem, the expected difficulty/payoff, and the citation. Call out which ideas are low-effort high-confidence wins vs research bets. Be explicit about anything unverified or license-incompatible.

Claim under review

"CausalTAD is a single-stage anchor-free temporal action detector built on ActionFormer, using a regression loss for boundary prediction and Soft-NMS post-processing rather than a dedicated over-segmentation/boundary-split mechanism."

Source: https://arxiv.org/html/2407.17792 (primary)
Supporting quote: "Our detection model is based on ActionFormer, a simple yet effective one-stage anchor-free method."

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

refuted=true if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.
refuted=false ONLY if: claim is well-supported, current, and source quality matches claim strength.
Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-06-16T03:25:58.530Z)

I'll verify this claim about CausalTAD by examining the primary source.

3. assistant (2026-06-16T03:26:01.166Z)

I'll fetch the primary source to verify the specific technical claims about CausalTAD.

4. user (2026-06-16T03:26:10.268Z)

CausalTAD Method Architecture

(1) Built on ActionFormer

Yes. The paper states: "Our detection model is based on ActionFormer, a simple yet effective one-stage anchor-free method." The key modification is replacing its transformer blocks with the proposed hybrid causal block.

(2) Single-Stage Anchor-Free

Confirmed. ActionFormer is described as "anchor-free," and CausalTAD inherits this property by design.

(3) Boundary Prediction Approach

The document mentions optimizing "the regression loss weight" as a hyperparameter but doesn't explicitly detail the regression mechanism. Standard anchor-free approaches predict boundary offsets, which appears applicable here.

(4) Post-Processing with Soft-NMS

Yes. For EPIC-Kitchens action detection, the method applies "Soft-NMS to eliminate redundant proposals." Additionally, "the sigma threshold in NMS during post-processing is set to 2.0 for the detection task."

(5) Over-Segmentation or Boundary-Split Mechanisms

No evidence of such mechanisms appears in the document.

(6) Novel Components Beyond ActionFormer

The primary innovation is the **hybrid causal block** combining two components:

- **Causal attention**: "restricting the context of attention to only past or future information" using masked self-attention
- **Causal Mamba**: "structured state-space model" adapted from VideoMambaSuite for bidirectional scanning

The paper emphasizes: "the model's access to only past or future context" mirrors "the natural progression of real-world events, the model can identify action transitions more accurately."

5. assistant (2026-06-16T03:26:12.430Z)

The fetch confirms the core architectural claims. Let me verify the "regression loss for boundary prediction" detail and check whether CausalTAD has any boundary/over-segmentation mechanism by searching independently.

6. user (2026-06-16T03:26:20.545Z)

Web search results for query: "CausalTAD temporal action detection ActionFormer causal attention Mamba EPIC-Kitchens 2024"

Links: [{"title":"Harnessing Temporal Causality for Advanced Temporal Action Detection","url":"https://arxiv.org/html/2407.17792v1"},{"title":"Harnessing Temporal Causality for Advanced Temporal Action Detection | Request PDF","url":"https://www.researchgate.net/publication/382560391_Harnessing_Temporal_Causality_for_Advanced_Temporal_Action_Detection"},{"title":"[PDF] Harnessing Temporal Causality for Advanced Temporal Action Detection | Semantic Scholar","url":"https://www.semanticscholar.org/paper/Harnessing-Temporal-Causality-for-Advanced-Temporal-Liu-Sui/2ed78841c110df5dc3515f9f8f8ec8087f8cbdb6"},{"title":"[Literature Review] Harnessing Temporal Causality for Advanced Temporal Action Detection","url":"https://www.themoonlight.io/en/review/harnessing-temporal-causality-for-advanced-temporal-action-detection"},{"title":"[PDF] End-to-End Temporal Action Detection with 1B Parameters Across 1000 Frames | Semantic Scholar","url":"https://www.semanticscholar.org/paper/End-to-End-Temporal-Action-Detection-with-1B-Across-Liu-Zhang/7149402c833dba1c073e999d0d47dbfa58a2b8a4"},{"title":"[2407.17792] Harnessing Temporal Causality for Advanced Temporal Action Detection","url":"https://arxiv.org/abs/2407.17792"},{"title":"Harnessing Temporal Causality for Advanced Temporal Action Detection - ????","url":"https://zhuanzhi.ai/paper/498f8e5e4d7b6dc2ccd842f382a48df4"},{"title":"Harnessing Temporal Causality for Advanced Temporal Action Detection","url":"https://arxiv.org/pdf/2407.17792"},{"title":"OpenTAD: A Unified Framework and Comprehensive Study of ...","url":"https://openaccess.thecvf.com/content/CVPR2025W/PVUW/papers/Liu_OpenTAD_A_Unified_Framework_and_Comprehensive_Study_of_Temporal_Action_CVPRW_2025_paper.pdf"},{"title":"Transformer or Mamba for Temporal Action Localization? Insights from","url":"https://www.scitepress.org/Papers/2025/131730/131730.pdf"}]

Based on the search results, here's what I found about CausalTAD:

Overview

CausalTAD combines causal attention and causal Mamba to achieve state-of-the-art performance on multiple benchmarks. It ranked 1st in the Action Recognition, Action Detection, and Audio-Based Interaction Detection tracks at the EPIC-Kitchens Challenge 2024, as well as 1st in the Moment Queries track at the Ego4D Challenge 2024.

Key Innovation

While existing modules typically treat past and future information equally, overlooking the crucial fact that changes in action boundaries are essentially causal events, the approach leverages the temporal causality of actions to enhance TAD representation by restricting the model's access to only past or future context.

Architecture

The detection model is based on ActionFormer, a one-stage anchor-free framework, and integrates the hybrid causal block in place of the traditional transformer block for improved contextual awareness during action detection. The synergy between causal attention and causal Mamba allows for more precise and contextually aware action detection, setting new benchmarks in the field.

Performance Results

The model achieves 34.99% average mAP on the test set of the Ego4D Moment Queries task and 31.97% average mAP on the test set of the EPIC-Kitchens 100 action detection task, ranking first in these tracks at the EgoVis Challenge 2024.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

7. assistant (2026-06-16T03:26:30.899Z)

Both sources corroborate the claim. Let me verify the specific "regression loss for boundary prediction" sub-claim, since ActionFormer is well-documented to use a regression head for boundary offsets. Let me confirm this is accurately characterized.

The claim has several components to assess against the evidence:

1. **"Single-stage anchor-free temporal action detector built on ActionFormer"** ? CONFIRMED. Direct quote: "Our detection model is based on ActionFormer, a simple yet effective one-stage anchor-free method." Both the abstract/search results and the paper itself confirm ActionFormer base, one-stage, anchor-free.

2. **"Using a regression loss for boundary prediction"** ? SUPPORTED but slightly inferential. The paper mentions tuning "the regression loss weight." ActionFormer (the documented base) uses

a DIoU/regression head predicting distances to action boundaries from each instant. This is accurate as a characterization of how ActionFormer-based detectors do boundary prediction.

3. **"Soft-NMS post-processing"** ? CONFIRMED. The paper applies "Soft-NMS to eliminate redundant proposals" with sigma 2.0.

4. **"Rather than a dedicated over-segmentation/boundary-split mechanism"** ? CONFIRMED. The novelty is the hybrid causal block (causal attention + causal Mamba), not any over-segmentation control or boundary-splitting module. No such mechanism is present. The claim correctly contrasts this with the over-merge problem framing in the research question.

The claim is accurate. The framing ("rather than a dedicated over-segmentation/boundary-split mechanism") is correct and contextually important ? CausalTAD is an actionness-detection-style model, not an over-segmentation-control method. The only minor weakness is that "regression loss for boundary prediction" is partly inferred from ActionFormer's known design rather than spelled out explicitly in CausalTAD's text, but this is a faithful, standard characterization, not an overreach.

8. user (2026-06-16T03:26:38.128Z)

Structured output provided successfully

9. assistant (2026-06-16T03:26:43.016Z)

Verified the claim against the primary source and corroborating references. The claim is well-supported and accurate.

Findings: refuted=false, confidence=high. All four components of the claim check out against the primary source (arXiv 2407.17792): (1) one-stage anchor-free on ActionFormer is verified verbatim; (2) regression loss for boundaries is supported (paper tunes "regression loss weight"; ActionFormer base uses a boundary-offset regression head); (3) Soft-NMS post-processing is verified (sigma 2.0); (4) the absence of any dedicated over-segmentation/boundary-split mechanism is confirmed ? the sole novelty is the hybrid causal block (causal attention + causal Mamba). The only minor caveat is that "regression loss for boundary prediction" is partly inferred from ActionFormer's known design rather than spelled out in CausalTAD's own text, but it remains a faithful characterization, not an overreach.